

Scaling Behavior of MVDR Beamforming in Fixed-Spacing Large Linear Microphone Arrays

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Abstract—Microphone array and beamforming technologies are widely used in speech acquisition and processing applications. To improve speech capture performance, large-scale microphone arrays have attracted increasing attention, as the achievable spatial gain generally increases with the number of microphones. However, a fundamental question is whether continuously increasing the number of microphones can lead to sustained and proportional performance improvements, or whether beyond a certain array size, adding more microphones results in only minor performance improvements. This paper investigates the performance gain of oracle near-field minimum variance distortionless response (MVDR) beamforming as the number of microphones increases in room acoustic environments. Uniform linear arrays with fixed inter-element spacing are considered, and performance gains normalized by the number of microphones are evaluated. Results show that beamforming performance consistently improves as the array size increases; however, the incremental performance gain obtained by adding more microphones gradually diminishes. This observation indicates that enlarging the array yields diminishing marginal returns while incurring higher acquisition and computational costs, thereby revealing practical scalability limitations of beamforming with large-scale microphone arrays.

Index Terms—Large-scale microphone arrays, MVDR beamformer, near-field beamforming.

I. INTRODUCTION

Microphone arrays, combined with beamforming techniques, play a central role in modern speech acquisition and enhancement systems by enabling interference and noise suppression while preserving the desired speech signal [1-3]. Among the various beamforming approaches developed in the literature [4-6], the minimum variance distortionless response (MVDR) beamformer has been extensively studied and widely adopted due to its ability to minimize the output noise power [7-10].

Since beamforming relies on spatial sampling of the sound field, its performance fundamentally depends on the number and configuration of microphones in the array. In theory, increasing the number of array elements should lead to higher array gain [11], [12]. This naturally motivates the use of large-scale microphone arrays with larger apertures and richer spatial sampling. Such arrays offer enhanced spatial resolution and

improved the signal-to-noise ratio (SNR), and have been successfully applied to tasks such as distant speech acquisition [13-15], sound source localization [16-18], and source directivity estimation [19].

However, most array signal processing methods, originally developed for small-scale arrays, do not readily scale to large configurations. Applying these methods to large-scale arrays often introduces new challenges. In particular, the computational complexity increases significantly due to high-dimensional covariance matrix estimation and inversion, while in adaptive beamforming, obtaining reliable second-order statistics becomes increasingly difficult in practical noisy and reverberant environments [2], [20], [21].

To address these issues, an effective strategy is to decompose a large-scale array into multiple virtual subarrays. By leveraging this structure, Kronecker beamformers can be designed to substantially reduce the number of free parameters and computational complexity [2], [20], [22], [23].

Despite these advances, a fundamental question remains: how does beamforming performance scale with the number of microphones, and does it continue to improve monotonically or eventually saturate as the array size increases? Understanding this scaling behavior is crucial for evaluating the cost-performance trade-off in large-scale microphone array systems.

To address the above questions, this paper presents an analysis of how the performance gain of oracle near-field MVDR beamforming evolves with the number of microphones in room acoustic environments. In particular, we focus on fixed-spacing uniform linear arrays (ULAs) to characterize the corresponding empirical performance scaling behavior. Although the analysis is conducted for ULAs, the observed diminishing-return trend reflects more general scalability characteristics of large-scale microphone arrays.

The number of microphones is successively doubled from 2 to 512, and the array gain is evaluated using multiple objective metrics. The results show that, while the array gain consistently increases with the number of microphones, the marginal performance improvement obtained by adding additional microphones gradually decreases. Moreover, the normalized performance gain per microphone also decreases as the number of sensors grows. This behavior indicates that adding more microphones yields progressively smaller perfor-

This work was supported in part by the National Natural Science Foundation (NSFC) of China under Grant 62471340 and 62192713, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2025A151010226.

mance improvements while incurring higher hardware costs and system complexity, thereby revealing inherent scalability limitations that must be carefully considered in the design of large-scale microphone array beamforming systems.

II. SIGNAL MODEL AND PROBLEM FORMULATION

Consider a ULA consisting of M omnidirectional microphones with an inter-element spacing of δ . Under a near-field acoustic propagation model, the propagation function from a source located at $\mathbf{r}_s = [r_{s,x} \ r_{s,y} \ r_{s,z}]^T$ (where T denotes the transpose operator) to the m th microphone positioned at $\mathbf{r}_m = [r_{m,x} \ r_{m,y} \ r_{m,z}]^T$, for $m = 1, 2, \dots, M$, can be expressed as [24]

$$G_{s,m}(f) = \frac{1}{4\pi\delta_{s,m}} e^{-j2\pi f\delta_{s,m}/c}, \quad (1)$$

where f denotes the temporal frequency, $\delta_{s,m} = \|\mathbf{r}_s - \mathbf{r}_m\|_2$ represents the distance between the source and the m th microphone, with $\|\cdot\|_2$ being the Euclidean norm, j is the imaginary unit, and c is the speed of sound in air.

In the frequency domain, the array observation vector $\mathbf{y}(f) \in \mathbb{C}^{M \times 1}$ can be written as [1]

$$\begin{aligned} \mathbf{y}(f) &= [Y_1(f) \ Y_2(f) \ \dots \ Y_M(f)]^T \\ &= \mathbf{d}_s(f) X(f) + \mathbf{v}(f), \end{aligned} \quad (2)$$

where $Y_m(f)$ is the observed signal at the m th microphone, with $m = 1, 2, \dots, M$, $X(f)$ denotes the desired signal at the reference microphone, $\mathbf{v}(f) \in \mathbb{C}^{M \times 1}$ is the additive noise vector defined analogously to $\mathbf{y}(f)$, and $\mathbf{d}_s(f) \in \mathbb{C}^{M \times 1}$ is the relative transfer function vector, defined as [24]

$$\mathbf{d}_s(f) = \begin{bmatrix} \frac{G_1(\omega, \delta_{s,1})}{G_{\text{ref}}(\omega, \delta_{s,\text{ref}})} & \frac{G_2(\omega, \delta_{s,2})}{G_{\text{ref}}(\omega, \delta_{s,\text{ref}})} & \dots & \frac{G_M(\omega, \delta_{s,M})}{G_{\text{ref}}(\omega, \delta_{s,\text{ref}})} \end{bmatrix}^T, \quad (3)$$

with $G_{\text{ref}}(\omega, \delta_{s,\text{ref}}) = G_1(\omega, \delta_{s,1})$ denoting the propagation function at the reference microphone, which is selected as the first sensor in this work.

Beamforming aims to estimate the desired signal by applying complex-valued weights to the outputs of each microphone and summing the weighted signals. Mathematically, this process can be expressed as [1], [6]

$$\begin{aligned} Z(\omega) &= \mathbf{h}^H(\omega) \mathbf{y}(\omega) \\ &= \mathbf{h}^H(\omega) \mathbf{d}_s(\omega) X(\omega) + \mathbf{h}^H(\omega) \mathbf{v}(\omega), \end{aligned} \quad (4)$$

where $\mathbf{h}(\omega) \in \mathbb{C}^{M \times 1}$ denotes the beamforming filter and H represents the Hermitian transpose. To ensure that the desired signal passes through the beamformer without distortion, a distortionless constraint is imposed in the desired direction:

$$\mathbf{h}^H(\omega) \mathbf{d}_s(\omega) = 1. \quad (5)$$

The task now is to determine the optimal beamformer $\mathbf{h}(\omega)$. A widely used solution is the MVDR beamformer, which is

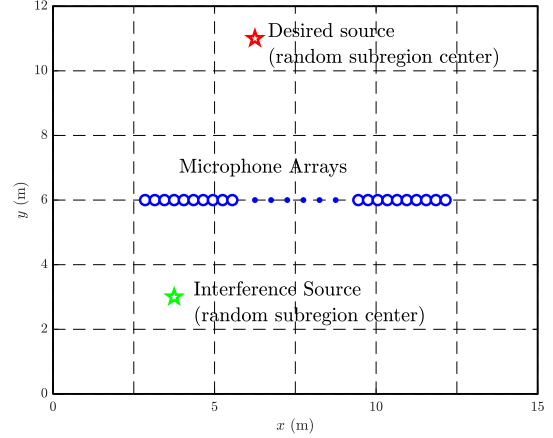


Fig. 1: Illustration of the simulated room acoustic environment, with the horizontal plane uniformly divided into 6×6 subregions.

obtained by minimizing the variance of the residual noise while satisfying the distortionless constraint [7]. The resulting MVDR beamformer is given by

$$\mathbf{h}_{\text{MVDR}}(\omega) = \frac{\Phi_{\mathbf{v}}^{-1}(\omega) \mathbf{d}_s(\omega)}{\mathbf{d}_s^H(\omega) \Phi_{\mathbf{v}}^{-1}(\omega) \mathbf{d}_s(\omega)}, \quad (6)$$

where $\Phi_{\mathbf{v}}(\omega) = E[\mathbf{v}(\omega) \mathbf{v}^H(\omega)] \in \mathbb{C}^{M \times M}$ is the covariance matrix of $\mathbf{v}(\omega)$, with $E[\cdot]$ denoting mathematical expectation.

III. SIMULATIONS

This section presents a validation study of the performance scaling behavior of MVDR beamforming in large-scale microphone array configurations, rather than introducing a new algorithm. By fixing the array geometry and progressively increasing the number of microphones, the experiments isolate the effect of array size on beamforming performance.

Specifically, simulations are conducted to quantitatively evaluate oracle near-field MVDR beamforming in room acoustic environments. In this setting, the source locations are assumed to be known and are used to compute the relative transfer function vector in (3), while the noise covariance matrices are also assumed to be available.

A. Simulation Configuration

To simulate reverberant room acoustic environments, impulse responses are generated using the image method [25]. An illustration of the simulated room setup is shown in Fig. 1. The room dimensions vary from $12 \times 12 \times 3$ m to $15 \times 15 \times 4$ m.

A linear array of omnidirectional microphones is placed at the center of the room, at a height between 1 and 2 m. The number of microphones is varied from 2 and increased exponentially up to 512, with an inter-element spacing of $\delta = 0.02$ m.

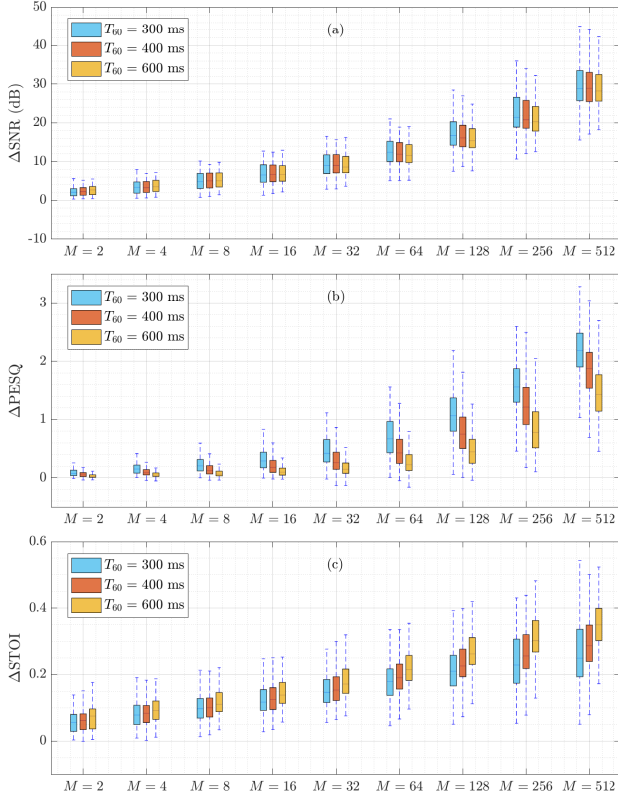


Fig. 2: Performance of the MVDR beamformer as a function of the number of microphones: (a) Δ SNR, (b) Δ PESQ, and (c) Δ STOI. Conditions: ULA and $\delta = 0.02$ m.

The room is uniformly divided into 6×6 subregions along its length and width on the horizontal plane. A desired source and an interfering source are randomly placed at the centers of the subregions, with an angular separation greater than 15° , and are assumed to be coplanar with the microphone array. The reverberation time T_{60} is set to three different levels: 300 ms, 400 ms, and 600 ms.

200 clean speech utterances are randomly selected from the test-clean subset of the LibriSpeech ASR corpus as target signals [26], while interfering noise is drawn from the DEMAND dataset [27]. All signals are resampled to 16 kHz.

The clean speech and interference signals are first convolved with their corresponding simulated room impulse responses and mixed at SNRs ranging from 0 to 10 dB. Subsequently, diffuse noise with SNRs between 10 and 20 dB and additive spatio-temporal white noise with SNRs between 20 and 30 dB are added, respectively.

To implement the frequency-domain MVDR beamformer, all signals are transformed and processed into the STFT domain. A frame length of 512 samples is used with a 75% overlap between consecutive frames, and the Hanning window is employed for both analysis and synthesis. The performance of the MVDR beamformer is evaluated using the following metrics: SNR, perceptual evaluation of speech quality (PESQ) [28], short-

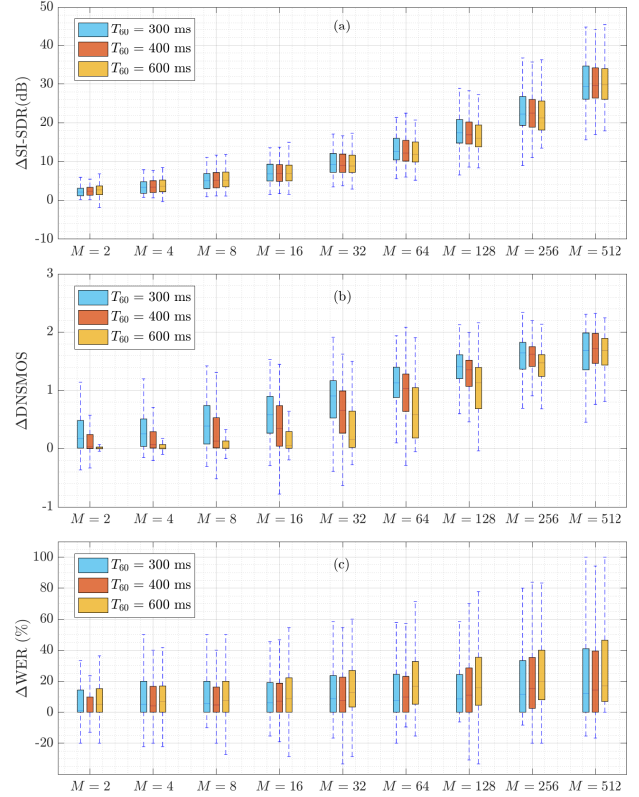


Fig. 3: Performance of the MVDR beamformer as a function of the number of microphones: (a) Δ SI-SDR, (b) Δ DNSMOS, and (c) Δ WER(%). Conditions: ULA and $\delta = 0.02$ m.

time objective intelligibility (STOI) [29], and scale-invariant signal-to-distortion ratio (SI-SDR) [30]. In addition, to reduce reliance on reference signals, two reference-free metrics: deep noise suppression mean opinion score (DNSMOS) [31] and word error rate (WER), are also included in the evaluation. To highlight performance improvements, all results are reported as gains relative to the metrics computed from the signal at the reference sensor (i.e., the first microphone in the array).

B. Simulation Results

Figures 2 and 3 show the performance of MVDR beamforming as a function of the number of microphones for fixed-spacing ULAs ($\delta = 0.02$ m) under the three specified reverberation conditions. The results consistently indicate that increasing the number of microphones improves performance across all scenarios; however, the gains do not scale proportionally with the number of sensors or the overall array size.

A closer examination reveals that when the number of sensors grows from very small configurations, performance improves rapidly, indicating that adding microphones effectively enhances the array gain. However, as the number of microphones continues to increase, the marginal benefit of adding more sensors gradually diminishes. For instance, increasing the number of microphones from 64 to 128 (an addition of 64

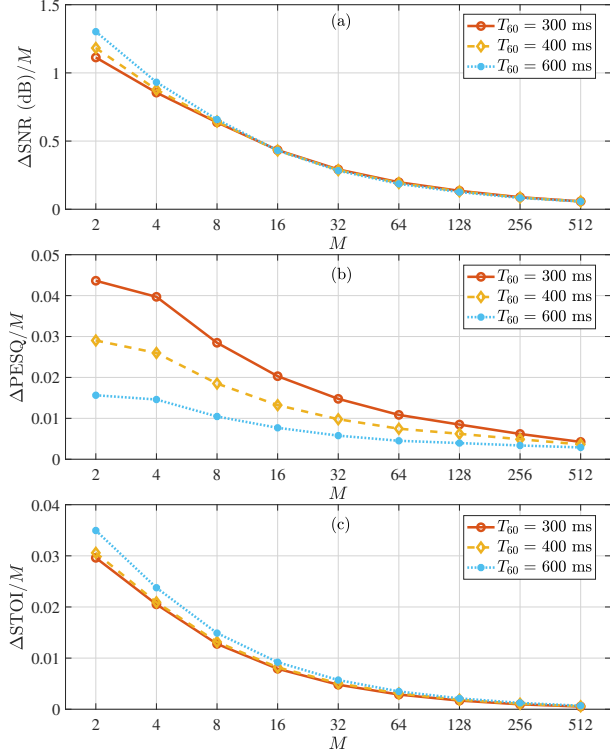


Fig. 4: Normalized performance gain of the MVDR beamformer with respect to the number of microphones: (a) $\Delta\text{SNR}/M$, (b) $\Delta\text{PESQ}/M$, and (c) $\Delta\text{STOI}/M$. Conditions: ULA and $\delta = 0.02$ m.

sensors) results in an approximate ΔSNR improvement of 5 dB, whereas expanding the array from 256 to 512 microphones (an addition of 256 sensors) yields only about a 7 dB gain. One possible explanation lies in the limited spatial coherence across microphones in reverberant environments. Partial correlation of noise and reverberation between channels, along with imperfect signal coherence across the array, can reduce the marginal performance gains as the array grows larger [11].

To further characterize the scaling behavior, Figs 4 and 5 present the normalized performance gains of MVDR as a function of the number of microphones. Specifically, the gains are normalized by the number of sensors, allowing the contribution of each individual microphone to be examined explicitly. This normalization reflects the average performance contribution per microphone and provides a clearer view of how the marginal utility evolves as the array grows. It can be observed that, across all evaluation metrics and reverberation conditions, the normalized performance gains decrease monotonically with the number of microphones. For very large arrays, the normalized gain approaches zero. This indicates that each additional microphone contributes less improvement than the previous one, i.e., the marginal benefit progressively diminishes, which is consistent with the diminishing-return behavior noted earlier.

At the same time, increasing the number of sensors inevitably

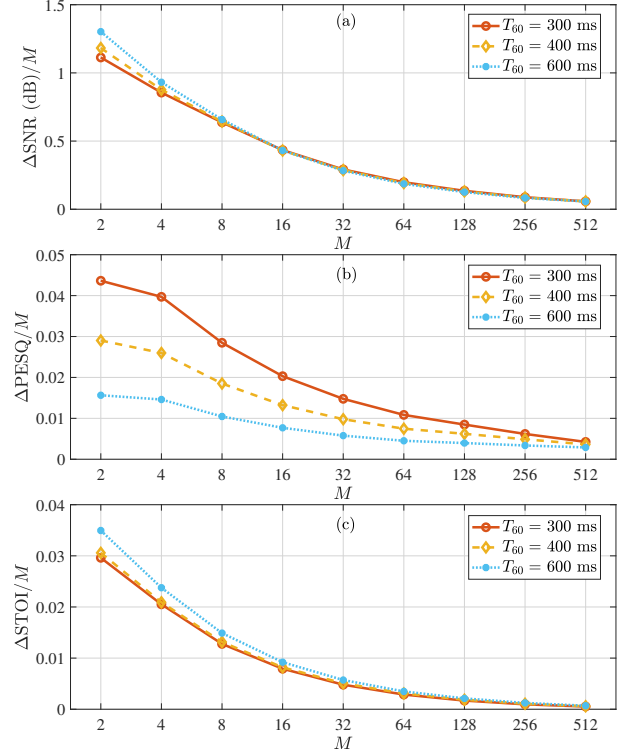


Fig. 5: Normalized performance gain of the MVDR beamformer with respect to the number of microphones: (a) $\Delta\text{SI-SDR}/M$, (b) $\Delta\text{DNSMOS}/M$, and (c) $\Delta\text{WER}(\%)/M$. Conditions: ULA and $\delta = 0.02$ m.

leads to higher hardware costs and system complexity, both of which typically grow rapidly with the number of microphones. These observations reinforce the conclusion that simply scaling up the array beyond a certain point is an inefficient strategy.

IV. CONCLUSIONS

This paper examined the scaling behavior of MVDR beamforming in fixed-spacing uniform linear microphone arrays under oracle near-field assumptions and room acoustic conditions. By progressively increasing the number of microphones, the effect of array size on beamforming performance was systematically evaluated using multiple objective metrics. The results show that although increasing the array size consistently improves MVDR beamforming performance across different reverberation conditions, the contribution of each additional microphone decreases monotonically as the array grows, revealing a clear diminishing-return effect. These findings demonstrate the inherent scalability limits of MVDR beamforming with fixed-spacing ULA and provide practical guidance for array design, highlighting the trade-off between achievable performance gains and the increased system complexity associated with larger arrays.

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